FYS4480/9480 November 5

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FCI

$$|\Psi_0\rangle = |\Phi_0\rangle + \sum_{pH} C_H^P |\Phi_H^P\rangle$$

$$C_H^P = C_{i_1 i_2 \ldots i_H}^{a_1 a_2 \ldots a_P}$$

$$|\Phi_H^P\rangle = a_{a_1}^\dagger a_{a_2}^\dagger \ldots a_{a_P}^\dagger a_{i_H} a_{i_{H-1}} \ldots a_{i_1} \times |\Phi_0\rangle$$

$$= (1 + \hat{C}) |\Phi_0\rangle$$

RSPT: $$|\Psi_0\rangle = \sum_{n=0}^{\infty} \left[R_0 (E_0 - \Delta E + H_1) \right]^n |\Phi_0\rangle$$

$$R_0 = \sum_{m \geq 1} \frac{|\Phi_m\rangle\langle\Phi_m|}{D_m}$$

$$D_m = E_0 - E_m$$

$$H_0 |\Phi_0\rangle = E_0 |\Phi_0\rangle$$

$$H_0 |\Phi_m\rangle = E_m |\Phi_m\rangle$$

$$\langle H_1 \rangle = \langle \Phi_0 | H_1 | \Phi_0 \rangle$$

$$|\Psi_0\rangle = |\Phi_c\rangle + \underbrace{R_0 H_1 |\Phi_c\rangle}_{\Omega^{(1)}}$$

$$+ \underbrace{\left(R_0 H_1 R_0 H_1 - \langle H_1 \rangle R_0^2 H_1 \right) |\Phi_0\rangle}_{\Omega^{(2)}}$$

$$+ \underbrace{\left[R_0 H_1 R_0 H_1 R_0 H_1 - \langle H_1 \rangle \left(R_0 H_1 R_0^2 H_1 + R_0^2 H_1 R_0 H_1 \right) + \langle H_1 \rangle^2 R_0^2 H_1 - \langle H_1 R_0 H_1 \rangle R_0^2 H_1 \right] \times |\Phi_c\rangle}_{\Omega^{(3)}}$$

$$+ \left(\Omega^{(4)} + \cdots \right) |\Phi_0\rangle$$

in a determinant basis

$$\langle \Phi_m | H_1 | \Phi_n \rangle = V_{mn}$$

$$|\psi^{(1)}\rangle = \Omega^{(1)} |\Phi_0\rangle$$

$$= R_0 H_1 |\Phi_0\rangle$$

$$(R_0 H_1 P) |\Phi_0\rangle$$

$$P = |\Phi_0\rangle\langle\Phi_0|$$

$$= \sum_{m \geq 1} \frac{|\Phi_m\rangle\langle\Phi_m|H_1|\Phi_0\rangle}{E_0 - E_m}$$

$$\langle \Phi_m | \psi^{(1)} \rangle = \delta_{mn} V_{nn} / D_n \qquad m \geq 1$$

$$\langle \underline{\Phi}_m | \psi^{(2)} \rangle =$$

$$\langle \underline{\Phi}_m | \Omega^{(2)} | \underline{\Phi}_0 \rangle$$

$$= \sum_e \frac{V_{me}}{\varepsilon_0 - \varepsilon_m} \frac{V_{e0}}{\varepsilon_0 - \varepsilon_e}$$

$$- V_{00} \frac{V_{m0}}{(\varepsilon_0 - \varepsilon_m)^2}$$

$$\Delta E = \sum_{i=1}^{\infty} \Delta E^{(i)}$$

$$\Delta E^{(1)} = \langle \Phi_0 | H_1 | \Phi_0 \rangle$$

$$\Delta E^{(2)} = \langle \Phi_0 | H_1 R^{(1)} | \Phi_0 \rangle$$

$$= \langle \Phi_0 | H_1 R_0 H_1 | \Phi_0 \rangle$$

$$= \langle \Phi_0 | H_1 \sum_{m>1} \frac{|\Phi_m\rangle\langle\Phi_m| H_1 |\Phi_0\rangle}{E_0 - E_m}$$

$$= \sum_{m>1} \frac{V_{0m} V_{m0}}{D_m} = \sum_m \frac{|V_{0m}|^2}{D_m}$$

$$\Delta E^{(3)} = \langle \Phi_0 | H_1 R^{(2)} | \Phi_0 \rangle$$

$$= \langle \Phi_0 | H_1 R_0 H_1 R_0 H_1 | \Phi_0 \rangle$$

$$- \langle H_1 \rangle \langle \Phi_0 | H_1 R_0^2 H_1 | \Phi_0 \rangle$$

$$= \sum_{mn} \frac{V_{0m} V_{mn} V_{n0}}{(E_0 - E_m)(E_0 - E_n)}$$

$$- V_{00} \sum_m \frac{V_{0m} V_{m0}}{D_m^2}$$

$$\Delta E^{(4)} = \sum_{mnl} \frac{V_{om} V_{mn} V_{nl} V_{l0}}{D_m D_n D_l}$$

$$- 2 V_{00} \sum_{mn} \frac{V_{om} V_{mn} V_{n0}}{D_m^2 D_n}$$

$$+ V_{00}^2 \sum_{m} \frac{V_{om} V_{m0}}{D_m^3}$$

$$- \Delta E^{(2)} \sum_{m} \frac{V_{om} V_{m0}}{D_m^2}$$

$$H = H_0 + H_1$$

$$= \sum_{pq} h_{pq}\, a_p^\dagger a_q + \frac{1}{4} \sum_{pqrs} \bar{v}_{pqrs}$$

$$\left(\langle p|h|q\rangle = h_{pq} \right)$$

$$\times\, a_p^\dagger a_q^\dagger a_s a_r$$

$$\bar{v}_{pqrs} = \langle pq|v|rs\rangle - \langle pq|v|sr\rangle$$

$$H_0 = \sum_p \varepsilon_p\, a_p^\dagger a_p = \sum_p h_{pp}\, a_p^\dagger a_p$$

$$u_{pq} = h_{pq} - \varepsilon_p \delta_{pq}$$

$$U_{pq} = \sum_{j \le F} \bar{v}_{pjqj} \qquad (\text{HF potential})$$

$$\tilde{f}_{pq} = u_{pq} + U_{pq}$$

normal-ordering

$$H_1 = \sum_{ij} u_{ij} + \sum_{pq} \tilde{f}_{pq} \{ a_p^\dagger a_q \} + \frac{1}{4} \sum_{pqrs} \bar{v}_{pqrs} \{ a_p^\dagger a_q^\dagger a_s a_r \}$$

$$\Delta E^{(2)} = \sum_{m>1} \langle \Phi_0 | H_1 | \Phi_m \rangle \langle \Phi_m | H_1 | \Phi_0 \rangle / D_m$$

$$\sum_{pq} \langle \Phi_0 | a_p^+ a_q | \Phi_m \rangle \tilde{f}_{pq}$$

$$|\Phi_m\rangle = \begin{cases} a_a^+ a_r |\Phi_0\rangle \\ a_a^+ a_b^+ a_j a_i |\Phi_0\rangle \end{cases}$$

$$\sum_{ia} \frac{|\tilde{f}_{ai}|^2}{D_r^a} \qquad D_r^a = \varepsilon_i - \varepsilon_a$$

$$\Delta E^{(2)} = \sum_{ai} \frac{|\tilde{f}_{ai}|^2}{D_i^a}$$

$$+ \frac{1}{4} \sum_{\substack{abij \\ ij}} \frac{|\bar{v}_{abij}|^2}{D_{ij}^{ab}}$$

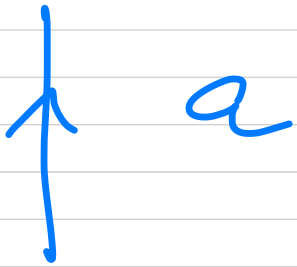
$$D_{ij}^{ab} = \varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b$$

$$\tilde{f}_{pq} = u_{pq} + U_{pq}$$

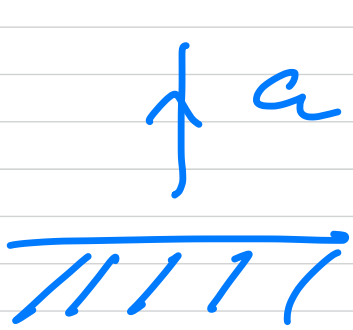
$$U_{pq} = \sum_{j \le F} \bar{v}_{pjqj}$$

Diagrammatic approach.

(i) particle



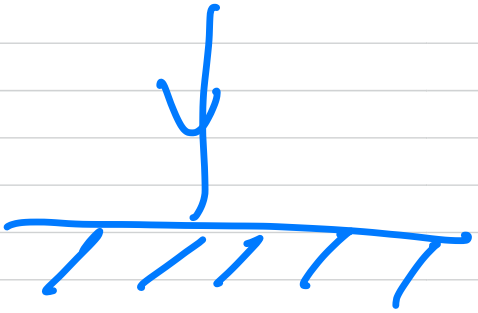
with vacuum



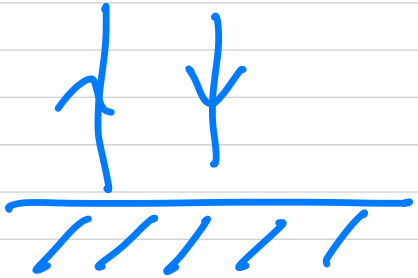
(ii) hole lines



with vacuum



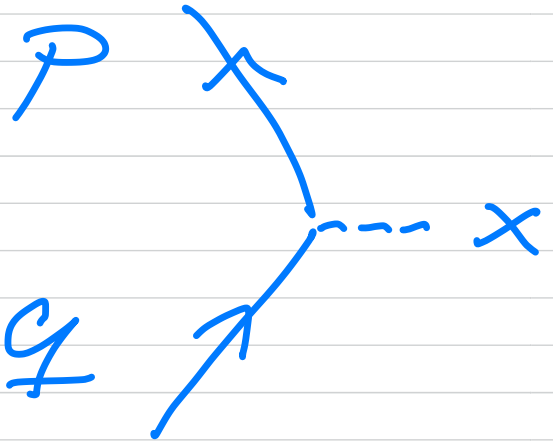
$a_i |\Phi_0\rangle$

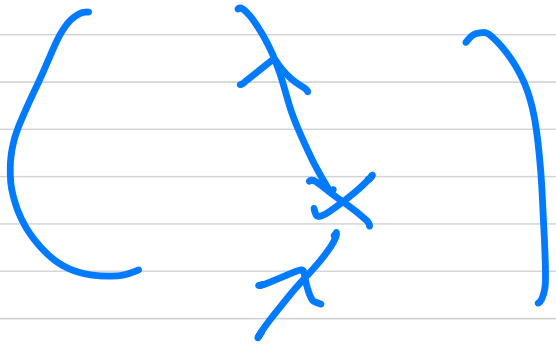


$a_a^\dagger a_i |\Phi_0\rangle$

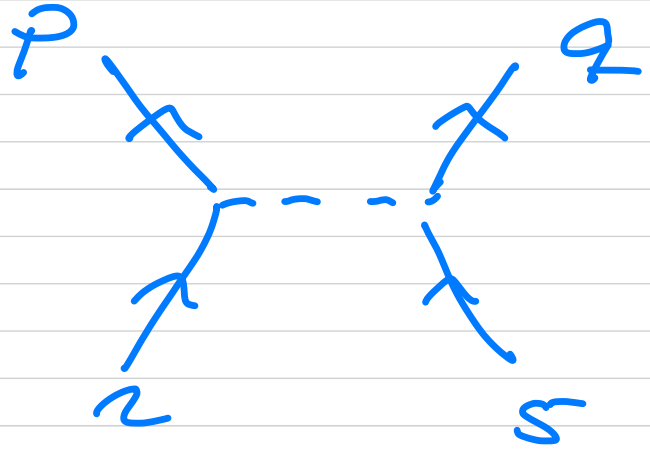
(iii)

$\tilde{\int}_{pq}$ or $\circ_{pq}$ (one-body)



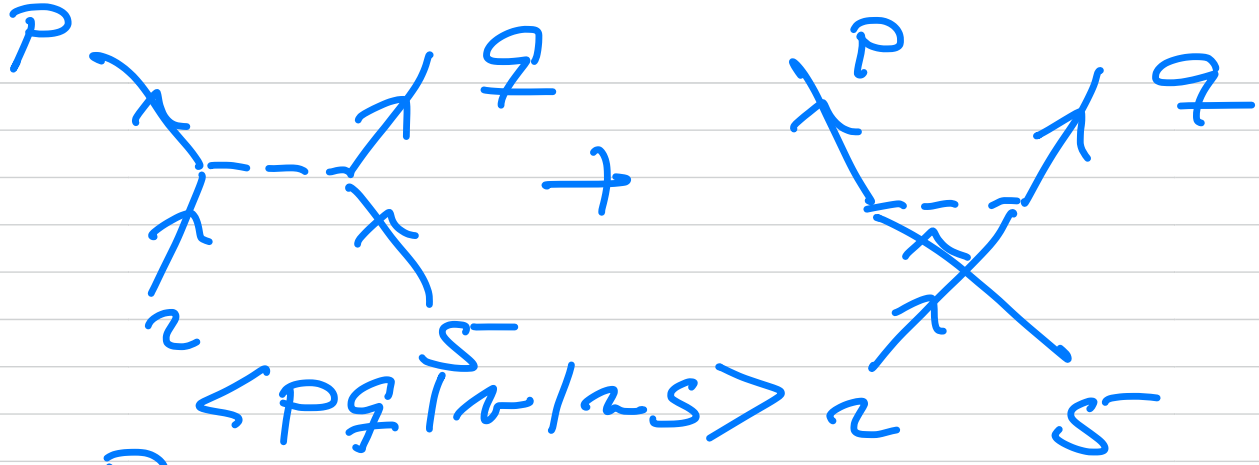


$\bar{v}_{pqrs}$ =



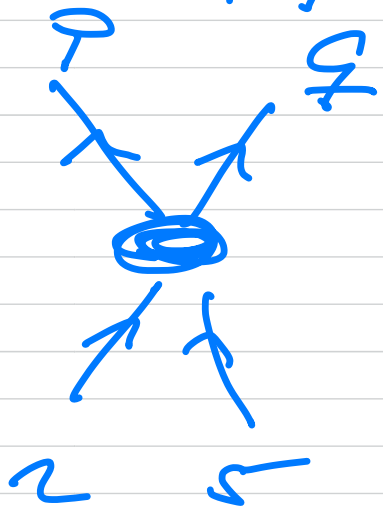
$\bar{v}_{pqrs} = \langle pq | v | rs \rangle - \langle pq | v | sr \rangle$

$\bar{v}_{pqrs}$ =



$\langle pq|v|rs\rangle$

$\bar{v}_{pqrs}$ =



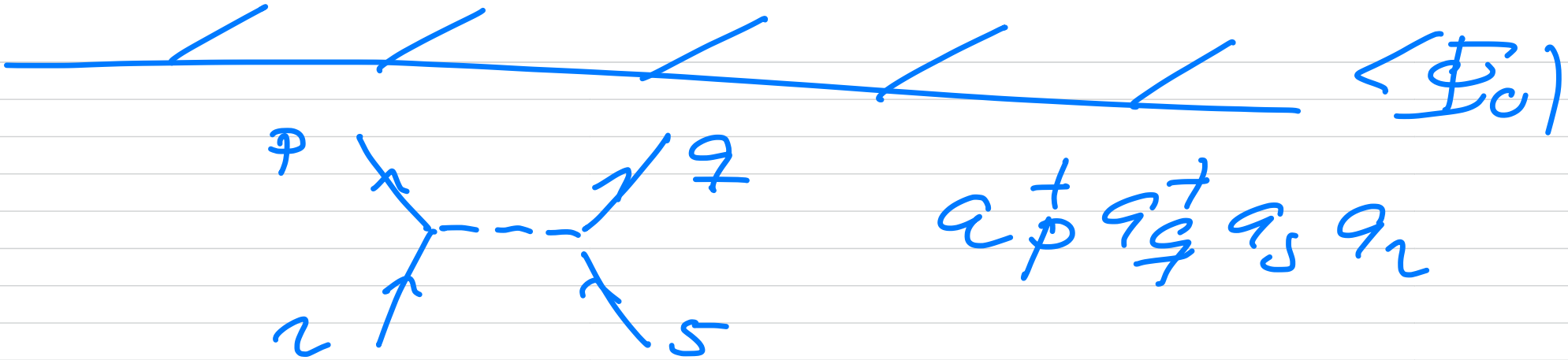
Hugenholtz diagram

$$\Delta E^{(2)} = \sum_m \frac{|V_{0m}|^2}{D_m}$$

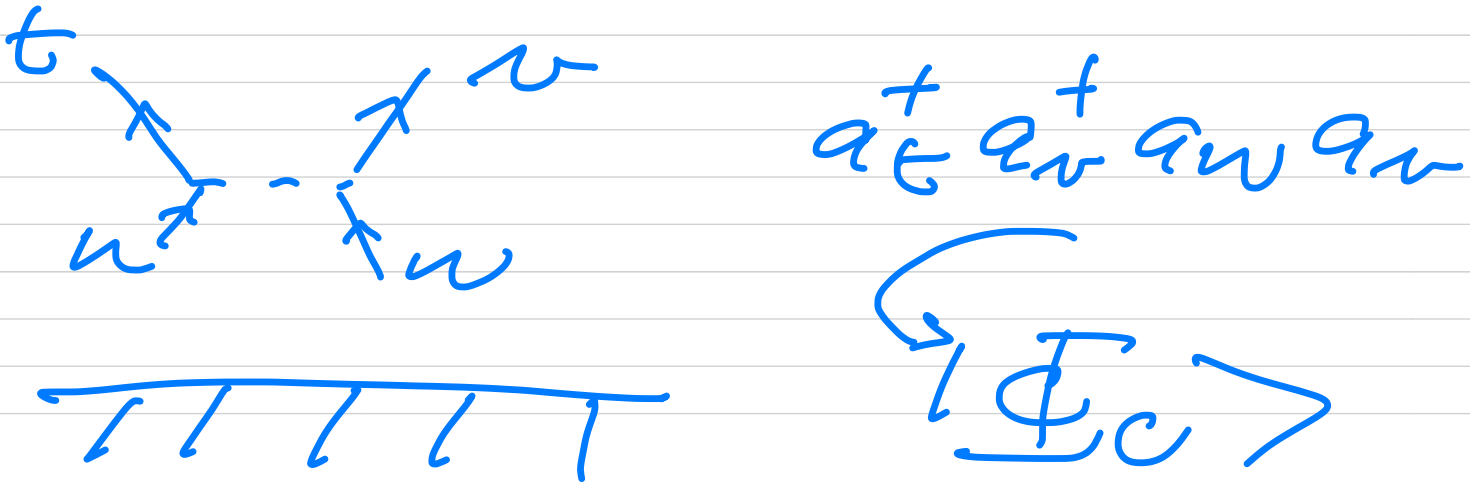
SD-basis

only $\frac{1}{4} \sum_{ab}^{rs} \frac{|\bar{v}_{ab}^{rs}|^2}{D_{ij}^{ab}}$

sp-basis



$$\sum_m \frac{|\Phi_m\rangle\langle\Phi_m|}{E_c - E_m}$$



$$\langle \Phi_m | a_t^\dagger a_\pi^\dagger a_u a_v | \Phi_c \rangle$$

$$\times \bar{v}_{tuvw}$$

$$v \; w \leq F$$

$$t \; u \leq F$$

$$v \; w \leq F \wedge t, u > F$$

$$(v \leq F \wedge t > F)$$

$\langle \Phi_m | \Phi_m \rangle \quad \bar{v}_{ab,ij}$

$\delta_{mm} \bar{v}_{ab,ij}$

$|\Phi_m\rangle = |\Phi_{ij}^{ab}\rangle$

next term in r

$D_{ij}^{ab} = \varepsilon_0 - \varepsilon_m =$

$\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b$

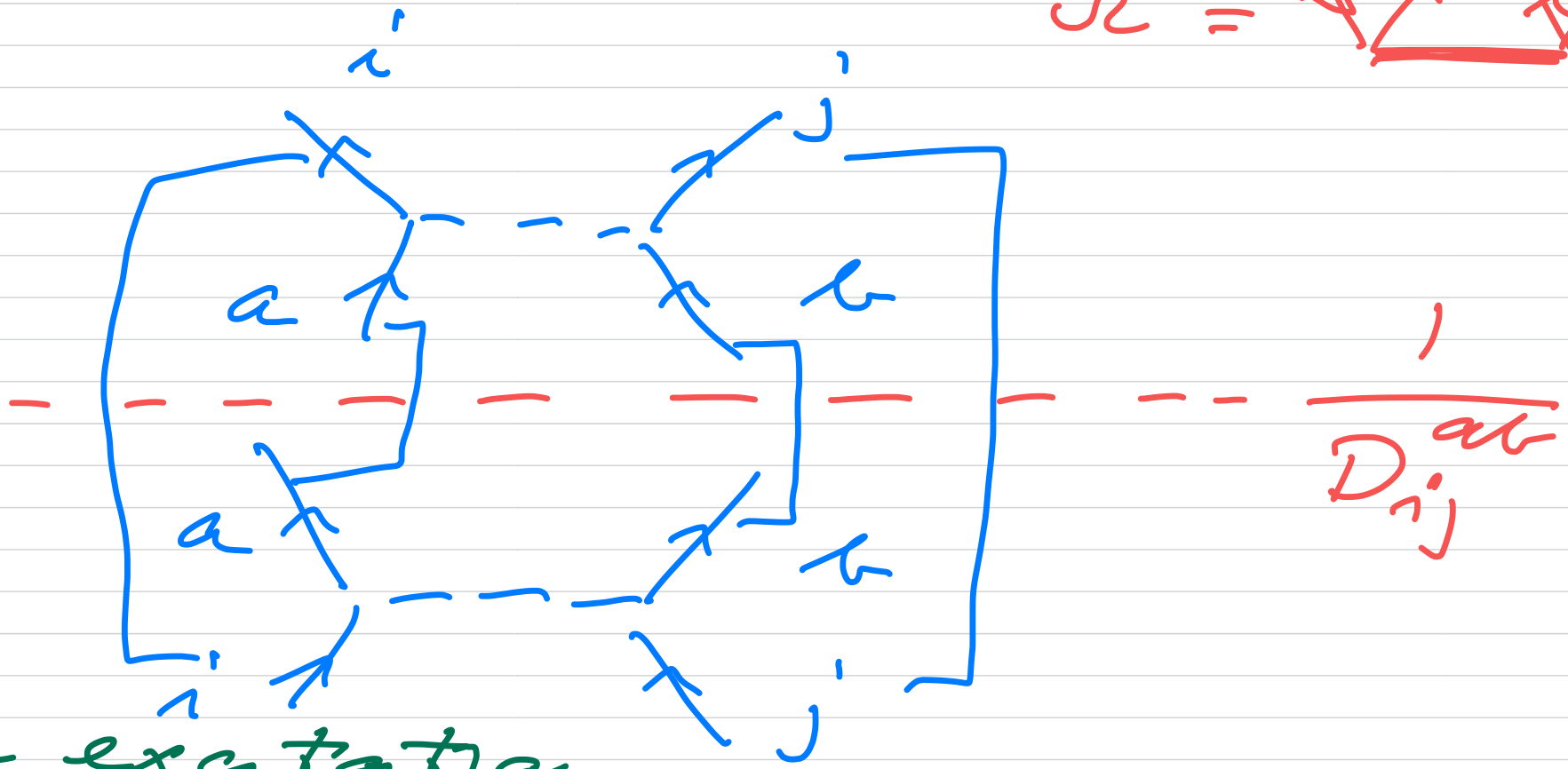
$\varepsilon_m = \varepsilon_0 + \varepsilon_a + \varepsilon_b - \varepsilon_i - \varepsilon_j$

$$\langle \Phi_0 | a_p^\dagger a_q^\dagger a_s a_r a_a^\dagger a_b^\dagger a_{i'}^\dagger a_{j'}^\dagger | \Phi_0 \rangle$$

$$\times \frac{\bar{v}_{ab}^{i'j'}}{\varepsilon_{i'} + \varepsilon_{j'} - \varepsilon_a - \varepsilon_b}$$

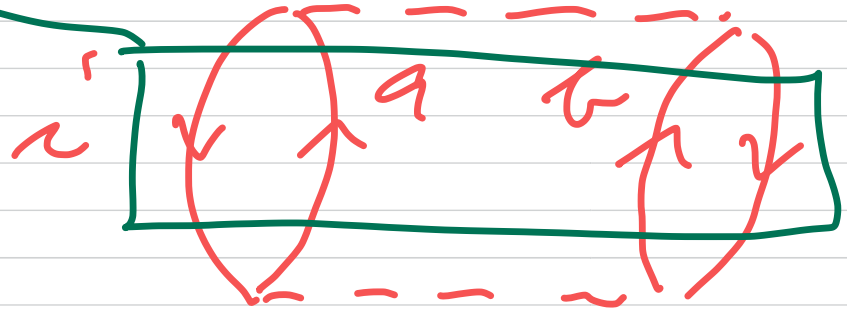
$$\Rightarrow \frac{1}{4} \sum_{\substack{ab \\ i'j'}} \frac{\bar{v}_{ab}^{i'j'}}{\varepsilon_{i'} + \varepsilon_{j'} - \varepsilon_a - \varepsilon_b}$$





$$\frac{1}{D_{ij}^{ab}}$$

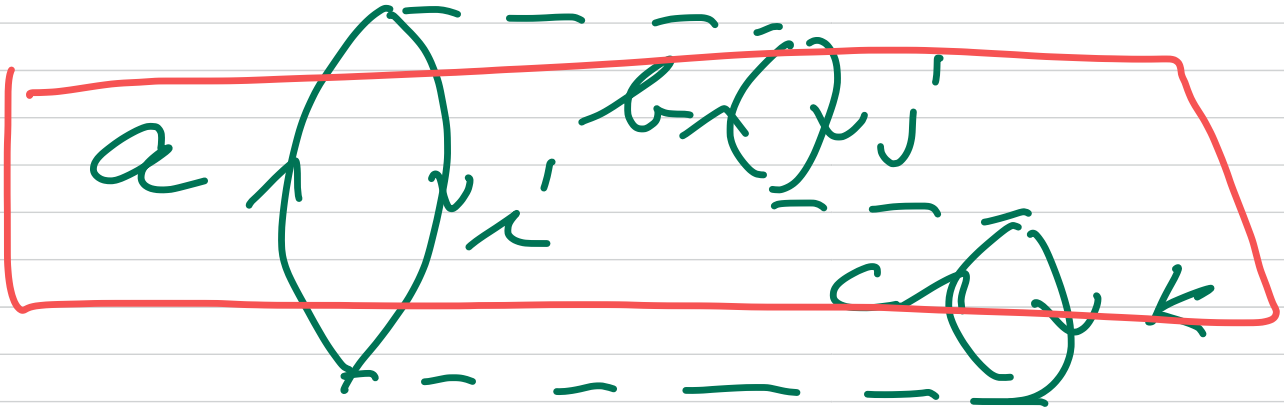
2p2h excitation



$$= \frac{1}{4} \sum_{ij}^{ab} \frac{|\bar{v}_{abij}|^2}{D_{ij}^{ab}}$$

$$= \langle \Phi_c | H_1 \Omega^{(1)} | \Phi_0 \rangle$$

3rd-order



– 3p3h excitation

Diagram rule

$$\left|\overline{\quad}\right|^2 \rightarrow \frac{1}{4} \sum_{\substack{ab \\ ij}} \frac{|\bar{V}_{ab,ij}|^2}{D_{ij}^{ab}}$$